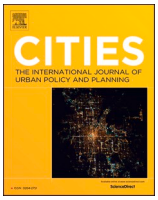
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Spatial city image and its formative factors: A street-based neighborhood cognition analysis

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ABSTRACT

This study analyzes the formative factors of the spatial image of two urban neighborhoods in Tokyo, Japan. Cognitive maps were collected from both residents and visitors, depicting spatial domains as imagined by the names of the neighborhoods. Using street segments as spatial units of analysis, this study analyzed the factors affecting individuals' perceptions of streets. As a result, as well as boundaries such as rivers and avenues, visual landscape features that were extracted from street images had non-trivial effects on street perception. In particular, the presence of visible streetscape greenery is positively associated with the probability of streets being perceived within the neighborhoods. However, the effect sizes differ between the two neighborhoods, suggesting the need for streetscape designs according to the existing city images. The results also revealed inter-individual heterogeneity in neighborhood cognition by gender, age, and familiarity with the neighborhood. While the original hypothesis was that the spatial extent of the perceived neighborhood would widen according to familiarity, the opposite result was obtained. This might explain the process of people obtaining certainty of neighborhood cognition and optimizing/fixing daily activity places according to experiences.

1. Introduction

Activity-travel behavior is based on spatial cognition. Humans acquire spatial city images through experiences in their environment, and this cognition in turn forms patterns of human activities in urban spaces (Downs & Stea, 1973; Kirk et al., 1963). As such, the image of a city mediates human-environment interaction (Lynch, 1960), and a better understanding of the formative factors of spatial city image can lead to the building of a better image. To this end, the seminal work of Lynch (1960) presented a framework for analyzing the relationship between the spatial cognition of a city and the external properties of the environment. Lynch summarized the five types of elements as representative formative factors of the cognitive map: paths, edges, nodes, districts, and landmarks. Although this theory has led and inspired research in spatial cognition, behavioral geography, and urban studies (e.g., Filomena et al., 2019; Golledge et al., 2000; McCunn & Gifford, 2018), the extent to which urban elements affect the spatial cognition of individuals is still unclear.

From a behavioral perspective, understanding the spatial domain perceived by people plays a key role in the planning of activities. Cognitive distance and cognitive accessibility (Canter & Tagg, 1975; Manley et al., 2021) define the spatial domain of physical activities, which can form a choice set (i.e., the set of activity places available to individuals) for travel and social behavior of people (Ben-Akiva et al., 1985). In the context of walkability, several studies have analyzed walking neighborhoods perceived by residents, based on cognitive maps drawn by them (e.g., Coulton et al., 2001; Smith et al., 2010). Guimpert and Hurtubia (2018) collected cognitive maps of residential neighborhoods in Santiago, Chile, converted them to mesh-based perception data, and modeled the mesh perception probability of a resident with discrete choice modeling. This quantitative approach successfully measured the effects of the land use attributes of the meshes and demographic characteristics of residents on the perception of walking neighborhoods.

Among elements of a city image, streets (paths) are considered dominant factors (Lynch, 1960), and the management and design of streets have attracted substantial attention in recent urbanism (e.g., Mueller et al., 2020). An extensive body of studies on physical activities (Brownson et al., 2009; Sung et al., 2013), walking route choice behavior (Basu & Sevtsuk, 2022; Oyama, 2023, 2024), and street-level perception (He et al., 2023; Ramírez et al., 2021; Wang et al., 2022) have suggested the importance of street quality. However, regarding

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spatial cognition, previous studies focused on the topology or geometry of street networks (Kim & Penn, 2004; Mohsenin & Sevtsuk, 2013). How street attributes (e.g., location or visual streetscape features) affect the spatial image of a city has not been explored sufficiently.

The objective of this study is to analyze the formative factors of the spatial image of two adjacent neighborhoods in Tokyo. While these neighborhoods do not have clear boundaries, they are known as popular neighborhoods where people live and visit for activities. We hypothesize that while people perceive it heterogeneously, the spatial image of a Tokyo neighborhood can be formed as a domain with similar common street characteristics, spreading around places that people daily use and that are named after the neighborhood (similarly to the concept of *district* of Lynch, 1960). To this end, we collect and analyze cognitive maps of the neighborhoods based on their names from both residents and visitors, using street segments as spatial units for the neighborhood cognition analysis (this paper calls this method a *street-based* analysis/modeling). In particular, focusing on the impact of visual landscape features of streets, we conducted street photography and extracted features from the images as pixel values. In addition to aggregate neighborhood cognition analysis, we formulated an individual-level street perception model as a function of street attributes and respondents' demographic characteristics, by using a discrete choice model. The model allows us to analyze the extent to which such attributes affect the perception probability of streets and to capture the individual heterogeneity in the perception. This quantitative analysis of formative factors of a spatial city image can contribute to the *image building* (such as refinement and branding) of neighborhoods and help the design/expansion of activity space.

The main difference to the studies on perceived walking neighborhoods (Guimpert & Hurtubia, 2018; Smith et al., 2010) is that we focus on urban areas and collect data from both residents and visitors, where the respondents identify their cognitive area based on the name of the neighborhood. Our analysis is based on street segments, instead of meshes as in Guimpert and Hurtubia (2018). Whereas meshes are spatial units arbitrarily defined by the analyst and not directly perceived by people, streets are public spaces where people travel or perform activities on a daily basis. This street-based modeling together with street image analysis allows us to capture the effect of visual street quality on perception.

2. Materials and methods

2.1. Study area

We selected for the study sites two adjacent urban neighborhoods in Tokyo, Japan, "Kiyosumi-Shirakawa" and "Kiba", which neighbor each other and are located 2–4 km from Tokyo Station (Fig. 1). Both neighborhoods have popular parks with a large amount of greenery: Kiyosumi Garden & Park in Kiyosumi-Shirakawa; Kiba Park in Kiba. Kiyosumi Garden is a historical Japanese landscape garden, originating from the residence of an Edo-era merchant, and Kiyosumi Park is a public park adjacent to the garden. Kiba Park is a metropolitan park with an area of 24.2 ha, with many family-friendly facilities such as sports and recreational facilities, a dog run, and children's play areas. As such, the two neighborhoods are popular both as residential areas and as places for daily visits, and people enjoy strolling there on weekends. Both neighborhoods are also composed of grid-shaped streets, and variations in spatial images can be expected between individuals. Several rivers and waterways flow through these areas, which can affect neighborhood cognition. In addition, targeting neighborhoods adjacent to each other allows us to analyze heterogeneity in cognition by neighborhood. For these reasons, these two neighborhoods were selected for the analysis, and there we conducted a neighborhood cognition survey of residents and visitors as well as a street landscape survey.

Note that because the Japanese address system is not street-based but neighborhood-based, the spatial domain of a neighborhood can be defined by a collection of residential addresses that includes the name of the neighborhood, which this study calls *address-based neighborhood*. The address-based neighborhood of Kiyosumi-Shirakawa includes addresses "XXX Kiyosumi, Koto Ward, Tokyo" and "XXX Shirakawa, Koto Ward, Tokyo", and that of Kiba includes "XXX Kiba, Koto Ward, Tokyo", wherein XXX is a number. They are shown by the red dotted boundaries in Fig. 2(a). However, an address-based neighborhood at the town level is unlikely to be clearly perceived by people because it does not have a visual/physical boundary or administrative function. Many guides to Tokyo for visitors or potential new residents indeed introduce the neighborhoods differently from (often more widely than) the address-based neighborhoods (e.g., Tokyo Cheapo, 2023; Tokyo Convention & Visitors Bureau, 2023). For example, the Museum of Contemporary Art Tokyo located in Kiba Park (the museum icon in Fig. 1) is often presented as an attraction of both Kiyosumi-Shirakawa and Kiba neighborhoods, and its address is in Miyosi, Koto Ward. A clear boundary is also not generally shown in such guides. Therefore, we hypothesized that people reveal heterogeneous cognition based on their experiences or prior information when imagining a neighborhood by its name.

2.2. Neighborhood cognition survey

To collect neighborhood cognition data, an on-street questionnaire survey was conducted at the star-marked point in Fig. 2(a) for each neighborhood, during three days of October 20, 21 and 23, 2020. The star-marked points were selected as the survey points considering that they are the closest locations to the metro station named after each neighborhood (Kiyosumi-Shirakawa Station of the Hanzomon Line, and Kiba Station of the Tozai Line) among the points along the main parks in the neighborhoods (Kiyosumi Garden & Park for Kiyosumi-Shirakawa and Kiba Park for Kiba). As such, these points are considered to be very likely to be included within the spatial image of the two neighborhoods, and we set them as *reference points*.

In the survey, the cognitive areas of the respondents were collected in the following way (also see Appendix E for the original questionnaire). The respondents identified their cognitive areas of the neighborhoods (i.e., the spatial domain they consider/imagine to be within the district on the basis of its name) by drawing a polygon on the basemap,[1] relying on their experiences or prior information (see Appendix A for examples). The basemap was a screenshot image from OpenStreetMap (OpenStreetMap contributors, 2017), printed at a scale of 1/10000 on A4-size paper. Note that we did not make any additions or emphases that may help the respondents indicate the address-based neighborhood. Respondents were also asked questions about demographic characteristics: age, sex, years of residence (for residents), and frequency of visits (for visitors). We used the years of residence and frequency of visits as the measures of familiarity with the neighborhoods, with the following hypothesis: Unfamiliar people tend to have a relatively narrow cognitive area (e.g., just around the well-known station and park) as they associate little with the neighborhood name; and the cognitive area gets wider with familiarity because daily life and visits bring more information of the neighborhood (e.g., activity places accessible from the neighborhood).

For each neighborhood, we collected cognitive maps from 25 residents and 25 visitors, resulting in 100 respondents in total from the two neighborhoods. Respondents who live within the areas shown on the basemap (not the address-based neighborhood) were defined as *residents* in this study. Fig. 2(b) shows the characteristics of the respondents, suggesting that the data were collected without significant bias regarding each demographic characteristic.

[1] The data collection by drawing is similar to Smith et al. (2010) and Guimpert and Hurtubia (2018), but this study is different to them because the cognitive maps were collected in urban areas, from both residents and visitors, by relying on the respndents' spatial images by neighborhood names.

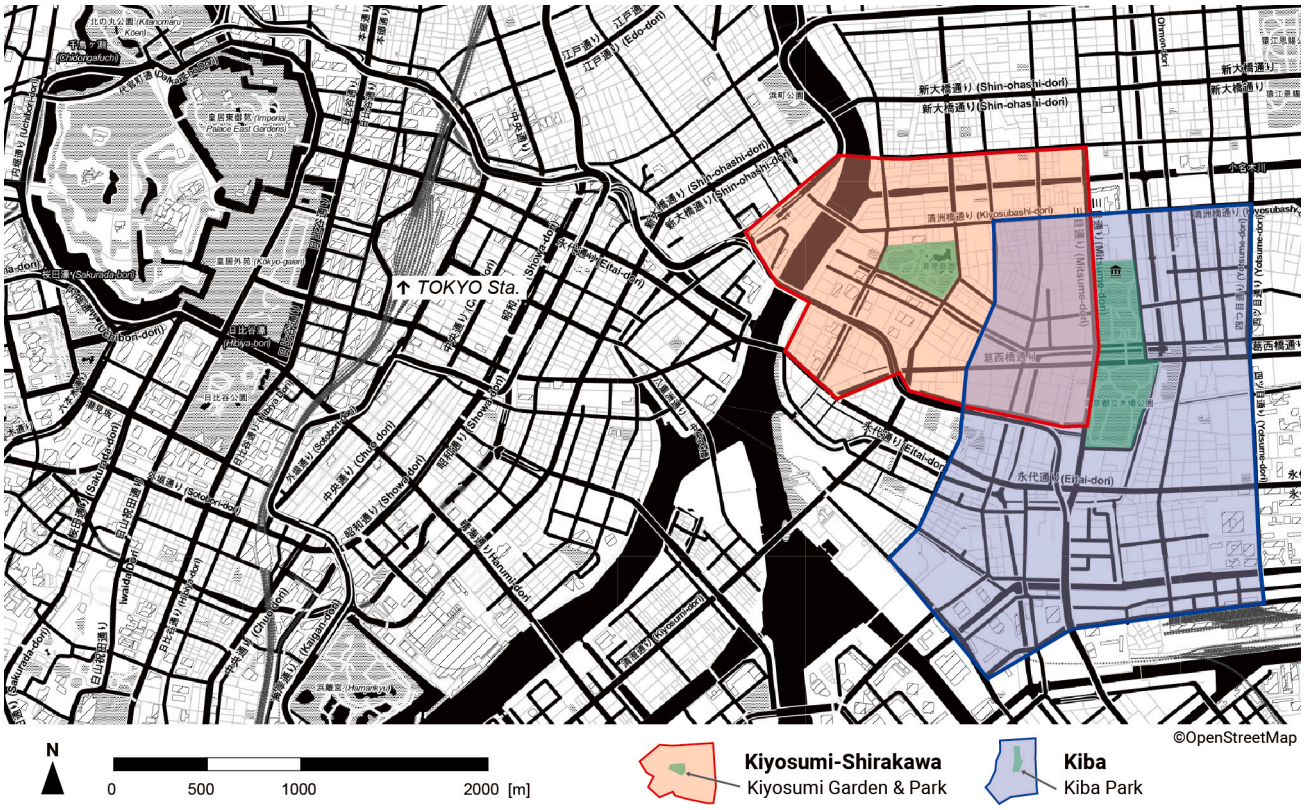


Fig. 1. Study sites: locations and surrounding environments of the two neighborhoods of Kiyosumi-Shirakawa and Kiba. (Source: OpenStreetMap).

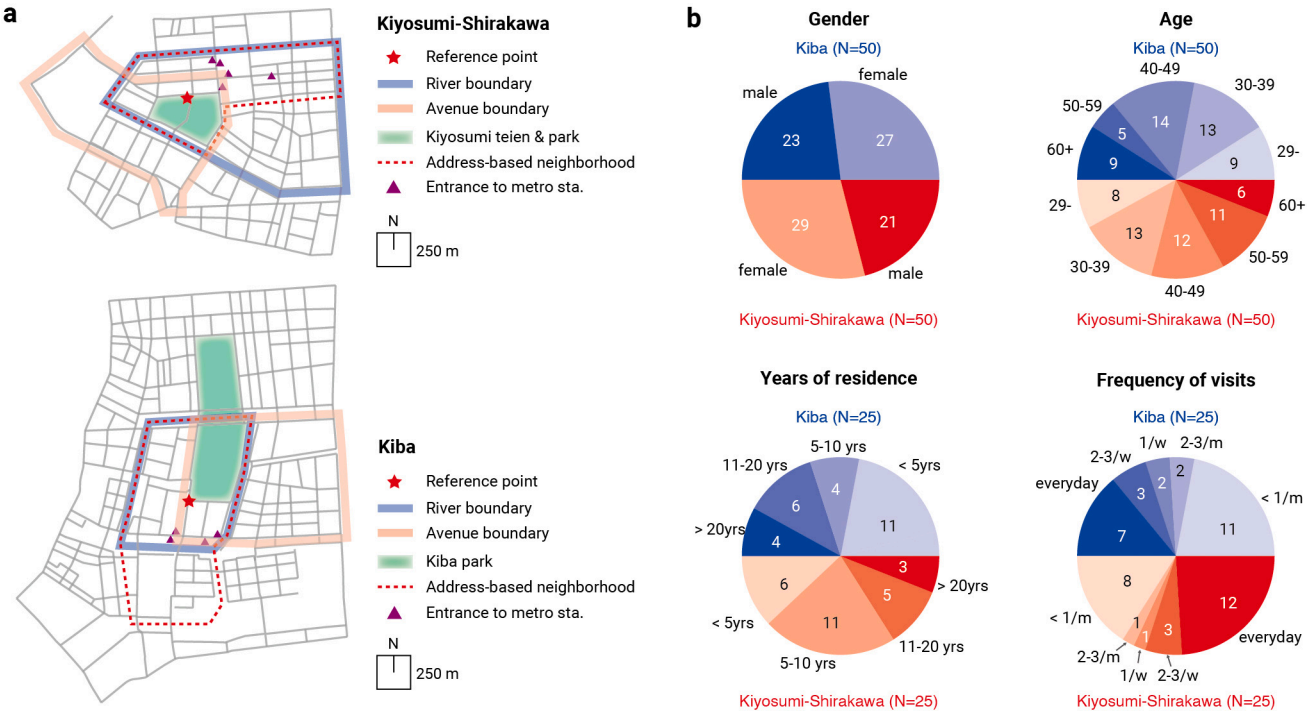


Fig. 2. (a) Survey networks: reference (survey) points, boundaries as well as parks and stations named after the districts; (b) Respondents' characteristics: gender, age, and familiarity with neighborhoods as measured by years of residence for residents and frequency of visits for visitors.

In addition, we converted the cognitive maps drawn by the respondents into street-based perception data $y_{ni} \in \{0,1\}$, indicating whether the respondent $n \in \{1,\ldots,N\}$ perceived the street $i \in \{1,\ldots,S\}$ within the neighborhood area ($y_{ni} = 1$) or not ($y_{ni} = 0$). Because the networks of Fig. 2(a) respectively include 258 (Kiyosumi-Shirakawa) and 473 (Kiba) streets, the numbers of observations account for 12,900 and 23,650, giving a total of 36,550. A street was defined as a section between two intersections, regardless of its length. In this sense, a street

in this study is different from a natural street and rather represents a street segment, but we use this definition hereafter. We also defined potential boundaries on the basis of the river/waterway and avenue closest to the reference point, as well as the address-based neighborhood, for each neighborhood. This data $\{y_{ni}\}$ is used directly for individual-based neighborhood cognition analysis. The observed perception rate for each street was also calculated as the average across the respondents, $\widetilde{p}_i = \sum_n y_{ni}/N$, which is used for the aggregate analysis.

2.3. Street photography and feature extraction

One of the main advantages of street-based neighborhood cognition analysis is the incorporation of street-based attributes, which people can perceive when traveling in the street network. In this study, a particular focus was placed on the visual landscape of streets, and its analysis is integrated with the neighborhood cognition analysis.

To this end, we took photos of all of the streets within the study networks (258 streets in Kiyosumi-Shirakawa, and 473 streets in Kiba) and collected street images. Although Google Street View images have often been used for street-level vision analysis in the literature (e.g., Nagata et al., 2020; Ramírez et al., 2021), we decided to take photos ourselves to control the photographic conditions. The photography was conducted over a five-day period from November 17 to 21, 2020, in the same season as the neighborhood cognition survey, during daylight hours from 9:00 a.m. to 4:00 p.m. Weather during the period was generally sunny and sometimes cloudy. We used an iPhone X camera set on a tripod and fixed it so that the height from the ground to the lens was 155 cm, an approximation of eye level. On each street, a single photo was taken from the point on the sidewalk (or roadside if there was no sidewalk) closest to the reference point. We carefully took the photo so that its area did not include adjacent streets or occupy a large portion of the close-range view. To bring the photo closer to the streetscape as seen by pedestrians, we shot toward the direction of travel from the entrance of the street. The vehicular and pedestrian traffic in the photos was also minimally controlled so that static landscape features could be extracted as accurately as possible.

Then, each street image was manually divided into five areas of Green, Building, Road, Sky and Other, and the area ratio (pixel) of each was calculated, using the Measurement Log function of Adobe Photoshop CC 2020. These pixel values are used as landscape features of streets.

2.4. Street-based neighborhood cognition model

We develop a street-based area cognition model based on data collected through the above-mentioned surveys. This model predicts the probability P_{ni} that an individual n perceives a street i within the area, formulated by a binary logit model:

$$P_{ni} \equiv P(y_{ni}=1) = \frac{1}{1+e^{-v_{ni}}}, \tag{1}$$

where v_{ni} is the deterministic component of utility for perception. This study formulates v_{ni} in a linear-in-parameter form:

$$v_{ni} = v_0 + \left(\beta_{\text{dist},0} + \sum_k \alpha_k s_{nk} + \sum_m \gamma_m \text{Direc}_{im}\right)\text{Dist}_i + \sum_l \beta_l x_{il}, \tag{2}$$

where v_0 is the intercept; Dist_i is the distance from the reference point to street i; and x_{il} are the other spatial attributes. The distance Dist_i is defined as the network shortest path distance from the reference point to the closer node of street i's end nodes.[2] To account for heterogeneity with respect to distance sensitivity, the coefficient for the distance variable Dist_i is defined as the sum of the base effect $\beta_{\text{dist},0}$, interactions with demographic characteristics $\mathbf{s}_n$ of individual n, and those with directions in which street i is located from the reference point Direc_i, which is a vector of four numeric values $(\text{East}_i,\ \text{North}_i,\ \text{South}_i,\ \text{West}_i)$ as defined in Appendix B. The parameter vector $\boldsymbol{\phi} = (v_0, \boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\gamma})$ is to be estimated from observations.

The parameter vector ϕ is estimated by solving the following likelihood maximization problem:

$$\max_{\boldsymbol{\phi}} LL(\boldsymbol{\phi}) = \sum_n \sum_i [y_{ni} ln P_{ni} + (1-y_{ni}) ln(1-P_{ni})] \tag{3}$$

We hereafter use a character modifier and write $\widehat{\boldsymbol{\phi}} = (\widehat{v}_0, \widehat{\alpha}, \widehat{\beta}, \widehat{\gamma})$ explicitly describing the estimated parameters.

With the estimated parameters, the model can evaluate policy impacts by calculating effect sizes or performing simulations (Parady & Axhausen, 2023). For discrete choice analysis, elasticity and marginal effect are used to measure the effect magnitude of a policy variable, defined as:

$$E_{x_{il}}^{P_{ni}} = \frac{\partial P_{ni}}{\partial x_{il}} \cdot \frac{x_{il}}{P_{ni}} = (1-P_{ni}) x_{il} \widehat{\beta}_l \tag{4}$$

$$M_{x_{il}}^{P_{ni}} = \frac{\partial P_{ni}}{\partial x_{il}} = P_{ni}(1-P_{ni}) \widehat{\beta}_l \tag{5}$$

Elasticity $E_{x_{il}}^{P_{ni}}$ measures the percentage change in the perception probability P_{ni} with respect to a given percentage change in an attribute x_{il}, while marginal effect measures the absolute change in P_{ni} with respect to a unit change x_{il}. This study focuses on the spatial attributes of streets and calculates elasticities for continuous variables and marginal effects for dummy variables.

3. Aggregate analysis

3.1. Distance and direction

We found that distance was one of the main effects: the further a street is from the reference point, the less it is perceived within the district. Fig. 3(a) shows the relationship between the observed perception rates of the streets and their distance from the reference point. Streets located more than 1 km from the reference point were rarely perceived (the means and 95 % confidence intervals of the perception rate are below 0.2). The decrease in the observed perception rates of the streets by distance is more rapid in Kiyosumi-Shirakawa than in Kiba. A possible explanation for this finding is related to the size of the main parks in the neighborhoods: Kiba Park, with an area of 24.2 ha, extends to the north and is much larger than Kiyosumi Garden & Park, with an area of 7.39 ha, suggesting that the presence of a large park may have extended the spatial image of the neighborhood.

To further investigate this result, we detailed the analysis by categorizing streets by directions. Fig. 3(b) and (c) show the results for Kiyosumi-Shirakawa and Kiba, respectively. In Kiyosumi-Shirakawa neighborhood, the streets in the east direction (colored green) are more likely to be perceived than those at the same distance in different directions. As Fig. 2(a) shows, the address-based neighborhood (red-dotted closed line) and the inner-river area (blue closed line) of Kiyosumi-Shirakawa extend to the east. The entrances to the Kiyosumi-Shirakawa station of the Tokyo Metro (purple triangles) are also located to the east of the reference point. In addition, popular cafes for young people and temples are distributed in the east area of the neighborhood,

[2] We considered this node as the access point of the street segment in the network, and this is consistent with the location where the street photo was taken.

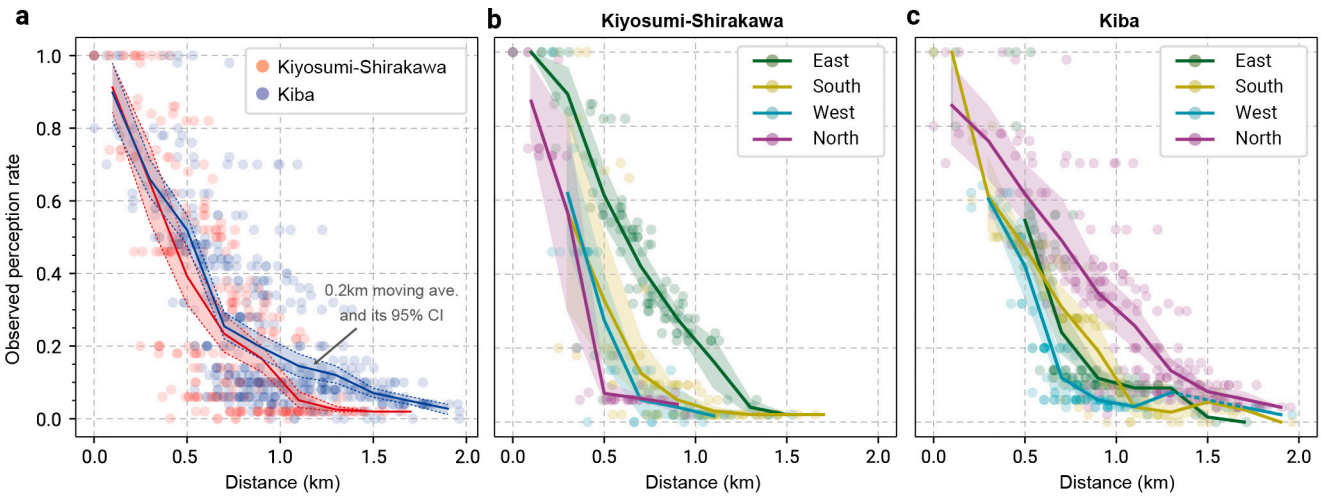


Fig. 3. Street perception and its distance from the reference point: (a) All street data, with red for Kiyosumi-Shirakawa, and blue for Kiba; (b)(c) Street data with differentiation by direction in Kiyosumi-Shirakawa and Kiba, respectively. Each dot in the plots indicates data for a single street, and the lines show the moving averages by 0.2 km along with the 95 % confidence intervals. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

which may be another reason why the cognitive area extended to the east. In Kiba neighborhood, the streets in the north had high observed perception rates. The main reason for this result seems to be the size and extent of Kiba Park, as discussed earlier. In fact, many respondents defined their cognitive area so that the entire of Kiba Park was included within it.

3.2. Boundaries

Another main factor defining the cognitive area is the existence of spatial boundaries. This study considered three different boundaries: river (waterway), avenue, and address-based neighborhood, as defined in Fig. 2(a). The box plots in Fig. 4 show the differences in street perception between inside and outside different boundaries. One-sided Mann-Whitney tests show that all boundaries have significant effects on street perception, with a confidence level of 99 %.

In particular, we found a clear effect of the river boundary in Kiyosumi-Shirakawa: the maximum observed perception rate of the streets outside the river boundary is approximately 0.2, and the effect size was 0.869,[3] suggesting that the river plays as a strong "edge" of the cognitive area of the neighborhood. The river also impacts the cognition of Kiba, but its effect size is smaller than in Kiyosumi-Shirakawa.

The effect of the existence of wide avenues is more ambiguous than the other boundary effects. Even outside the avenue boundary, there are streets with high perception rates in both neighborhoods. The address-based neighborhood also has a large effect on neighborhood cognition, but the variability of street perception rates outside the boundary is greater than that in the case of the river.

It should be noted that these results include the effect of distance because the streets inside the boundaries are closer to the reference point. Therefore, we show the model results in Section 4 to simultaneously consider and analyze in detail the effects of distance and boundaries.

3.3. Familiarity with neighborhoods

Finally, we focus on the heterogeneity among the respondents, with the hypothesis that familiarity with the neighborhoods has an impact on the extent and form of neighborhood cognition. Specifically, we hypothesize that the more familiar the respondents are with the neighborhood, the more information they have, and hence the wider their cognitive areas. To test this hypothesis, our survey collected information about the year of residence and the frequency of visits, as measures of familiarity with the neighborhoods.

Fig. 5(a) shows maps of the street perception rates of residents with different levels of residential years: short (lived for less than 5 years), long (more than 10 years), and medium-term (in-between). Interestingly, the results suggest the opposite tendency to our hypothesis. In both neighborhoods, the cognitive area appears to shrink as the years of residence in the neighborhood becomes longer. However, this change in the form of the cognitive area may be explainable, because in each neighborhood the cognitive area of residents who lived for more than 10 years seems to be similar to the shape of the address-based neighborhood. This pattern is particularly clear in Kiyosumi-Shirakawa, while the cognitive area in Kiba is also affected by the Kiba park. These results suggest that residents who lived in the area for a long time tend to conform their perceptions to the address-based neighborhood, whereas those who have recently started living there do not.

Fig. 5(b) shows the results of visitors with different levels of the frequency of visits: low (visit once a month or less), high (everyday), and medium (in-between). In contrast to the results of residents, the maps of visitors do not clearly show that people with a high level of familiarity with (i.e., who frequently visit) the neighborhood accurately perceive its area as the address-based neighborhood. Whereas the cognitive area of visitors seems to be similar to that of residents and extends to the south in Kiba, the cognitive area of visitors differs to that of residents in Kiyosumi-Shirakawa. The change in the cognitive area of visitors between low and medium frequencies indicates a similar trend to the case of residents, in the sense that people with low familiarity have a wide cognitive area, possibly because of ambiguous perception. However, the cognitive area of visitors with a high frequency of visits appears to expand again. A possible explanation for this result is that the cognitive area of visitors may indicate a spatial sphere for their possible activities, and its change in the result may be explained as the search and optimization process of the activity sphere. Visitors with wide and ambiguous neighborhood cognition may search and take a first visit to places in the neighborhood, then gradually optimize the places for their activities; however, after a certain point, some visitors may restart the search.

[3] The effect size r is the absolute value of z-score $|z|$ divided by the square root of the street number $\sqrt{S}$. As is commonly done, this study assumes the effect is small when $r < 0.3$, medium when $0.3 \leq r < 0.5$, and large when $r \geq 0.5$.

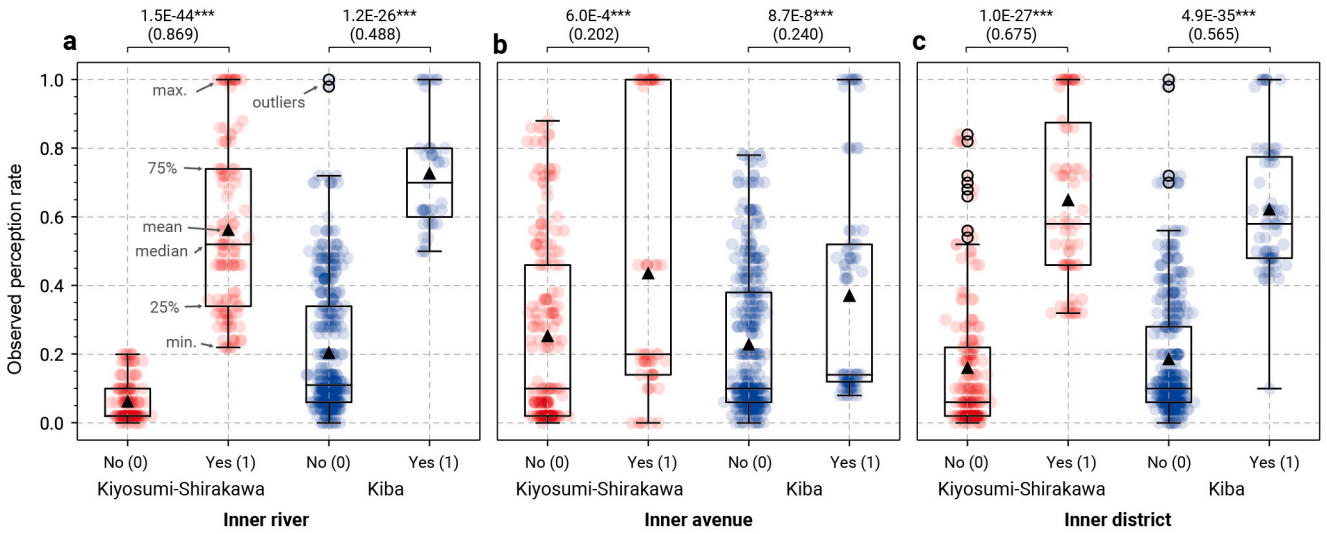


Fig. 4. Street perception and boundaries in box plots: with respect to (a) river, (b) avenue, and (c) address-based neighborhood boundaries. Each dot indicates data for a single street, with red for Kiyosumi-Shirakawa, and blue for Kiba. The numbers above the box plots are the *p*-values and effect sizes from one-sided Mann-Whitney tests, with stars indicating statistical significance. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

3.4. Streetscape clustering

Using the five types of pixel values of Green, Building, Road, Sky, and Other, extracted from street photos, we performed a cluster analysis of the streetscapes. The clustering was implemented by Lloyd's k-means algorithm (Lloyd, 1982). After some trials, we found that four was the optimal number of clusters for our analysis in terms of interpretation. The clustering results and the mean feature values (with mean observed perception rate of the streets) for each cluster are shown in Fig. 6(a) and (b), respectively. Although we separately performed the cluster analysis for the street data from Kiyosumi-Shirakawa and Kiba neighborhoods, the obtained results are similar to each other. As shown in Fig. 6(c), Green and Building pixel values are the main determinants of the clustering: Clusters A1 and B1 include streets with a lot of visible green; the streets in Clusters A2 and B2 have buildings as the dominant streetscape features and the least amount of visible green; the streets in Clusters A4 and B4 have the highest Road and Sky pixel values, implying a wide road width; and the streets in Clusters A3 and B3 have in-between features. We refer the reader to Appendix C for examples of street images for different clusters.

In both areas, Cluster 1 with a high value of Green pixels has the highest observed perception rate among the clusters, followed by Cluster 4, suggesting that the presence of visible green increases the probability of a street being perceived as included within the neighborhood.

Fig. 6(d) and (e) show how the streets in each cluster are distributed in the areas. In the Kiyosumi-Shiarakwa neighborhood, the streets in Clusters A2 and A3 that have high values of Building pixels are distributed in the eastern and southern areas in which many residential houses and shops are distributed. The streets in Clusters A1 and A4 are located around the Kiyosumi Garden & Park and along rivers/waterways. In the Kiba neighborhood, the streets in Clusters B1 and B4 that have relatively high values of Green pixels are distributed in the whole area, not only the surroundings of Kiba Park, and the streets in Cluster B2 with a high value of Building pixels can be seen in the western area.

4. Model analysis

This section discusses the estimation results of the street-based neighborhood cognition model defined in Section 2.4, and evaluates the extent to which some factors affect the street perception probability. The effect sizes were calculated according to Eqs. (4) and (5), and we report their averages over individuals and streets. The confidence intervals for elasticities and marginal effects were estimated via bootstrapping with 100 iterations (e.g., Parady et al., 2023; Parady & Axhausen, 2023). The model estimation and analysis were performed by *PandasBiogeme*, an open-source Python package for discrete choice analysis (Bierlaire, 2003, 2020).

We first discuss the specification search process, comparing different streetscape feature inputs, and focusing on heterogeneity with respect to neighborhoods and directions. We then show the estimation results and several scenario evaluations. Table 1 summarizes the definition and statistics of the explanatory variables used in the models.

4.1. Model comparison

The final model was reached after an extensive search for specifications. Table 2 reports the comparison of the models tested during the important stages of the specification search. We first focused on how to incorporate streetscape features into the model, and compared three different inputs for streetscape features, namely pixel values (Model 1), dissimilarity (Model 2), and cluster types (Model 3). Among them, Model 1 yield the best fit in terms of log-likelihood ($LL(\widehat{\phi})$), adjusted rho-squared ($\bar{\rho}^2$), as well as Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC). Importantly, because the pixel values are continuous variables, they enable policy analysis and provide easier interpretation, compared with the other inputs for streetscape features. The uses of the clustering result and dissimilarity have another limitation in that they depend on the definition of neighborhood areas (i.e., the set of streets included in the study areas). We thus conclude that pixel values are the best input of streetscape features for a street perception model.

We then estimated the model for each neighborhood separately to capture the heterogeneity of street perception mechanisms between the two neighborhoods (Model 4). This segmentation significantly improved the goodness-of-fit of the model, as suggested by the result of the likelihood ratio test against Model 1. We also validated the spatial transferability of the model and found that there was a significant difference in perception behavior between Kiyosumi-Shirakawa and Kiba (see

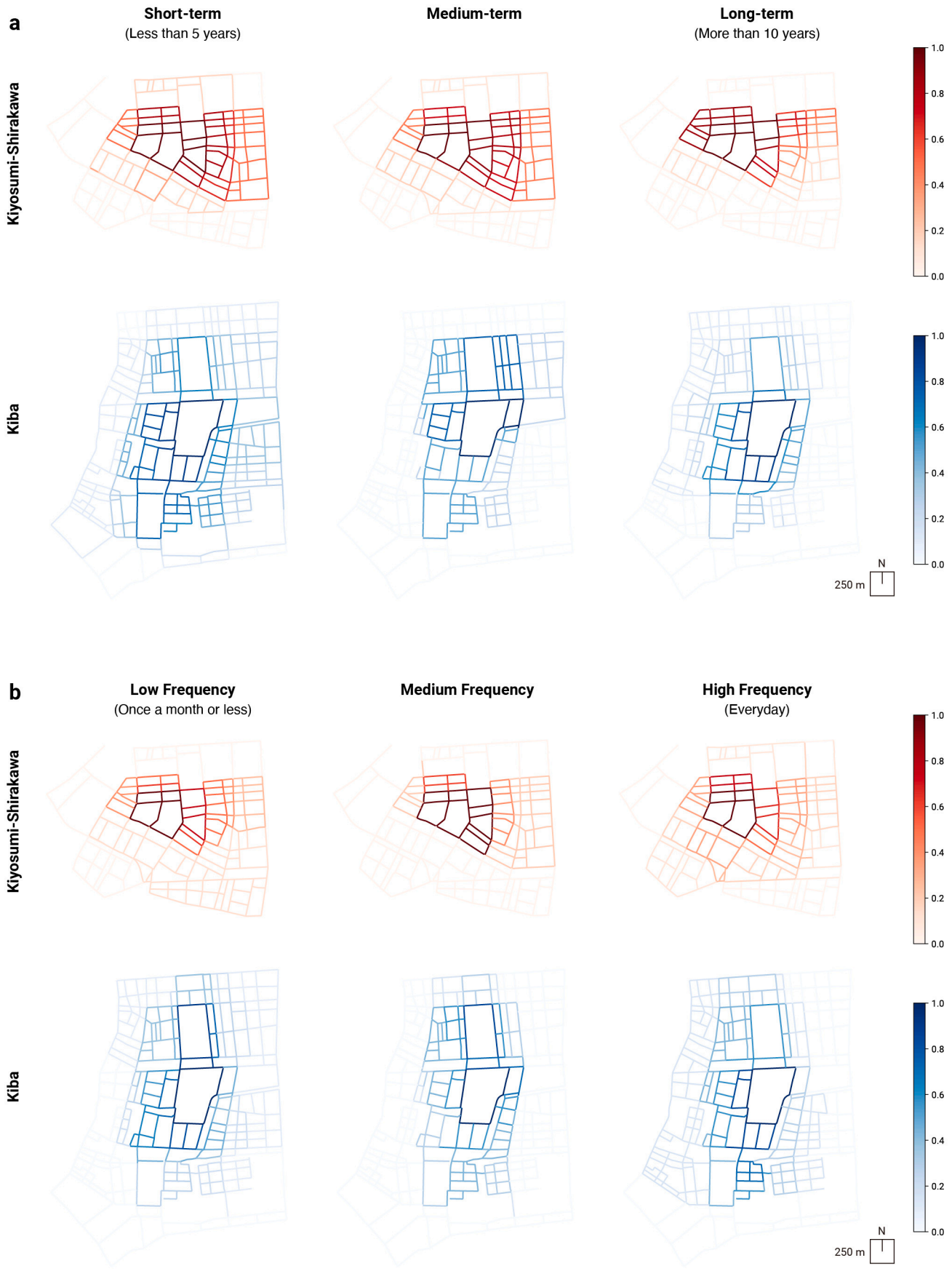


Fig. 5. Street perception and familiarity: mapping of observed perception rates of streets for (a) residents differentiated by years of residence; and for (b) visitors by frequency of visits. Red and blue plots represent Kiyosumi-Shirakawa and Kiba, respectively, and deeper colors indicate higher perception rates. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

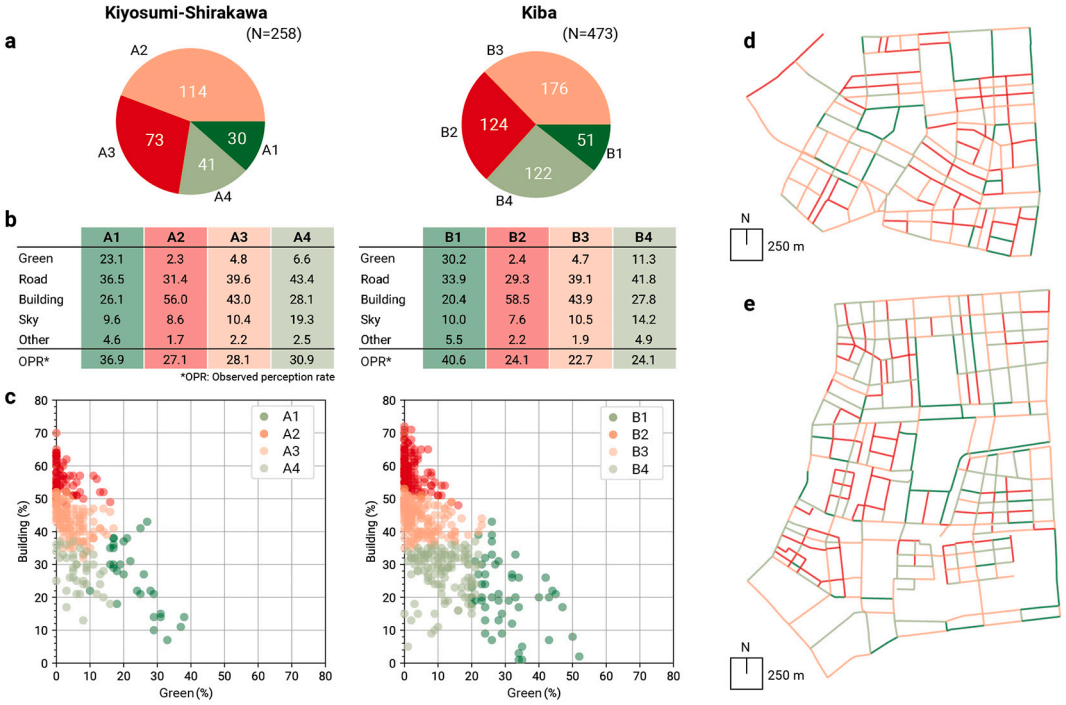


Fig. 6. Streetscape clustering result: (a) Number of streets in each cluster; (b) Average pixel values and observed perception rate of each cluster; (c) Plots of green and building pixel values of the streets; (d)(e) Mapping of the streets with cluster colors for Kiyosumi-Shirakawa and Kiba, respectively. For (a-c), the results of Kiyosumi-Shirakawa and Kiba are placed in the left and right columns, respectively. The colors in (d) and (e) correspond to those in (a-c). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Appendix D for more details).

The differentiation of the sensitivity to distance by directions in which the streets are located from the reference point also allows the model to better fit the observations,[4] indicated by the comparison between Models 4 and 5. The goodness-of-fit measures of Model 5 are all better than those of Model 4, and the likelihood ratio test also suggests a preference for Model 5 with a confidence level of 99 %.

As a result, we conclude that Model 5 is the best of all of the models considered. In the final model, the streetscape features are captured directly by pixels, and the sensitivity to distance is interacted with the demographic characteristics of the respondents and the directions of the streets. The final model is also estimated with segmentation by neighborhood; that is, the different parameters are estimated for the two neighborhoods of Kiyosumi-Shirakawa and Kiba.

4.2. Estimation results

Table 3 enumerates the estimation results of Models 1–3, and Table 4 reports the result of Model 5. The result of Model 4 was omitted for the sake of brevity. In all models, the estimated parameters have the expected signs, and most of them are statistically different from zero at the 1 % significance level.

As mentioned earlier, Models 1–3 were estimated with the composite sample and differently captured the effect of the streetscape. Model 1 suggests that the presence and visibility of green and sky may increase the probability of a street being perceived as within the areas of the two neighborhoods, while streets where buildings and roads are dominant features of streetscape are less perceived. Similarly, Model 3 suggests that the streets in Cluster 1 with a high value of Green pixels are more likely to be perceived as within the area, and those in Clusters 2 and 3 with a high value of Building pixels are less perceived. The positive association of street greenery and negative association of buildings with perception are consistent with the results of previous studies (e.g., Wang et al., 2022). In addition, the coefficient of dissimilarity of Model 2 has a negative sign, implying that streets with a similar streetscape to the neighborhood average are more likely to be included in the cognitive area.

For brevity, we henceforth focus on the estimation results of Model 5 (the final model), discussing heterogeneous distance sensitivities and the effects of boundaries and streetscape features.

4.2.1. Distance sensitivity

As expected, the base distance coefficient has a negative sign in both neighborhoods: the more distant a street is from the reference point, the less it is perceived to be within the neighborhood area. Women and people 40 years or older are in particular more sensitive to distance, meaning that their cognitive areas tend to be smaller than those of men and young people. These results correspond to previous results reported in the literature (e.g., Guimpert & Hurtubia, 2018). Note that the difference between the gender groups was not significant in Kiyosumi-Shirakawa.

Regarding familiarity with the neighborhoods (i.e., the residential years and visit frequency), we obtained an interesting result that is consistent with the analysis in Section 3.3. The coefficients of the familiarity measures interacting with distance were both estimated to be positive. This implies that people who are less familiar with the neighborhoods (residents who lived there for less than 10 years and visitors who visit there once a month or less) perceived the neighborhood area as being greater than those who are more familiar. This trend was consistent between the two neighborhoods.

The results also suggest clear directional heterogeneity of the sensitivity to distance. A positive value of the coefficient for a direction interaction term mitigates the resistance to distance. In Kiyosumi-

[4] Because the direction effects seemed to differ between the two neighborhoods, as suggested by the aggregate results in Section 3.1, we incorporated this heterogeneity after the neighborhood segmentation.

Table 1
Definition of explanatory variables. The statistics for street variables are for 258 and 473 streets in Kiyosumi-Shirakawa and Kiba, respectively. The stastistics for demographic characteristics are based on 50 respondents for each neighborhood.

			Kiyosumi-Shirakawa				Kiba			
Variable	Definition	Unit	Mean	Std.	Min	Max	Mean	Std.	Min	Max
Distance	Network shortest path distance from the observation point to the nearer of the two vertices of the street	Kilometer	0.73	0.33	0	1.66	0.93	0.39	0	1.96
Boundaries										
Inner river	Dummy variable. Equals 1 if the street is inside the river boundary, 0 otherwise.	Dummy	0.46	0.50	0	1	0.10	0.29	0	1
Inner avenue	Dummy variable. Equals 1 if the street is inside the avenue boundary, 0 otherwise.	Dummy	0.22	0.41	0	1	0.18	0.39	0	1
Inner district	Dummy variable. Takes value 1 if the street is inside the address-based neighborhood of the neighborhood, 0 otherwise.	Dummy	0.27	0.45	0	1	0.16	0.36	0	1
Streetscape features										
Building pixel	Pixel value of Building in the street image	Pixel	6.50	7.61	0	38	41.05	14.52	1	72
Road pixel	Pixel value of Road in the street image	Pixel	42.33	12.17	7	70	36.65	7.09	3	55
Sky pixel	Pixel value of Sky in the street image	Pixel	37.53	6.08	16	50	10.64	6.01	0	46
Green pixel	Pixel value of Green in the street image	Pixel	11.22	5.65	0	28	8.54	9.84	0	52
Other pixel	Pixel value of Other in the street image, which was treated as reference and not incorporated into the model because the sum of five pixels always equals one.	Pixel	2.41	3.49	0	27	3.12	4.82	0	48
Dissimilarity	Euclidean distance between streetscapes pixel values of the street of interest and of the neighborhood average.	Euclidean distance	29.93	9.49	3.98	54.22	35.83	12.77	5.83	64.78
Cluster 1	Dummy variable. Equals 1 if the street is in cluster 1, 0 otherwise.	Dummy	0.12	0.32	0	1	0.11	0.31	0	1
Cluster 2	Dummy variable. Equals 1 if the street is in cluster 2, 0 otherwise.	Dummy	0.28	0.45	0	1	0.26	0.44	0	1
Cluster 3	Dummy variable. Equals 1 if the street is in cluster 3, 0 otherwise.	Dummy	0.44	0.50	0	1	0.37	0.48	0	1
Cluster 4	Dummy variable. Equals 1 if the street is in cluster 4, 0 otherwise. This is treated as reference in the model.	Dummy	0.16	0.37	0	1	0.26	0.44	0	1
Directions (see Appendix B for the detail)										
East	Ratio of east direction in which the street is located	Ratio	0.36	0.35	0	0.997	0.25	0.32	0	0.994
North	Ratio of north direction in which the street is located	Ratio	0.15	0.29	0	0.990	0.35	0.37	0	0.999
South	Ratio of south direction in which the street is located	Ratio	0.32	0.30	0	0.960	0.19	0.27	0	0.954
West	Ratio of west direction in which the street is located, treated as reference in the model	Ratio	0.17	0.29	0	0.998	0.21	0.29	0	0.995
Demographic characteristics										
40 yrs or older	Dummy variable. Equals 1 if the respondent is 40 years or older, 0 otherwise.	Dummy	0.42	0.49	0	1	0.44	0.50	0	1
Female	Dummy variable. Equals 1 if the respondent is female, 0 otherwise.	Dummy	0.58	0.49	0	1	0.54	0.50	0	1
Live less than 10 yrs	Dummy variable. Equals 1 if the respondent is a resident who lived in the neighborhood for less than 10 years.	Dummy	0.34	0.47	0	1	0.30	0.46	0	1
Visit once a month or less	Dummy variable. Equals 1 if the respondent is a visitor who visits the neighborhood once a month or less.	Dummy	0.18	0.38	0	1	0.26	0.44	0	1

Shirakawa, the streets in the south are the least sensitive to distance, followed by the east, north, and west. Although the aggregate result of Fig. 3(b) suggested that the streets in the east were the least sensitive to distance, the result was based on discrete categorization and might not have captured well the difference between the north-east and south-east, and the use of continuous direction variables in the model captured it more flexibly. In Kiba, the streets in the north are the least sensitive to distance, followed by the south, east, and west. Consistent with the result of Fig. 3(c), this result captures the effect of the extent of Kiba Park.

4.2.2. *Effects of boundaries*

Next, we discuss the impacts of boundaries on the perception probability of streets. The positive signs of the coefficients for the inner-boundaries variables indicate that the boundaries played as edges of the cognitive area of the individuals. The effects were further examined by calculating their marginal effects, as reported in Table 5. The results suggest that the river boundary is the strongest edge of neighborhood cognition in both neighborhoods. It is particularly significant in Kiyosumi-Shirakawa, where on average a 28.1 % change in perception probability is suggested between inside and outside the river boundary, while the change is 11.0 % in Kiba. The avenue boundary has less impact on perception probability than the other boundaries in both neighborhoods. Nevertheless, all dummy variables of the inner boundaries are positively associated with street perception probability, indicated by the positive confidence intervals of their marginal effects.

4.2.3. *Effects of visual streetscape features*

Finally, we evaluated the extent to which the visual landscape features of the streets affect their perception probability in each

Table 2
Model comparison results. Likelihood ratio tests were performed to test statistical preference of Model 4 over Models 1, and of Model 5 over Models 4, for the pairs of hierarchically nested models.

	Model 1	Model 2	Model 3	Model 4	Model 5
Specifications					
Streetscape features input	Pixel	Dissimilarity	Cluster	Pixel	Pixel
Estimation by neighborhood[a]	x	x	x	✓	✓
Direction heterogeneity[b]	x	x	x	x	✓
Results					
Number of parameters[c]	13	10	12	26	32
$LL(\widehat{\phi})$	−16,338.250	−16,490.760	−16,415.203	−15,678.969	−14,969.240
$\bar{\rho}^2$	0.355	0.349	0.352	0.381	0.408
AIC	32,702.504	33,001.518	32,854.406	31,409.938	30,002.480
BIC	32,813.088	33,086.583	32,956.483	31,631.105	30,084.492
Likelihood ratio test					
w.r.t.				Model 1	Model 4
χ^2				1318.562	1419.458
d.f.				13	6
Preferred model				Model 4	Model 5
p				≤0.01	≤0.01

[a] Different parameters are estimated for the two neighborhoods.
[b] The interactions between distance and direction variables are included.
[c] This number includes the intercept.

Table 3
Estimation results of Model 1–3 with different landscape feature inputs.

	Model 1		Model 2		Model 3	
	Estimate	*t*-test	Estimate	t-test	Estimate	t-test
Intercept	0.898	2.94***	0.410	5.09***	0.387	7.01***
Distance (km)	−2.119	−32.72***	−2.085	−29.15***	−2.106	−30.86***
Distance × demographic chars.						
40 yrs or older	−0.278	−7.42***	−0.278	−7.53***	−0.277	−6.95***
Female	−0.475	−12.92***	−0.476	−12.71***	−0.474	−12.46***
Live less than 10 yrs	0.745	18.02***	0.742	16.51***	0.744	17.77***
Visit once a month or less	0.358	7.48***	0.363	5.90***	0.359	7.19***
Boundaries						
Inner river	1.090	30.95***	1.068	29.50***	1.109	31.69***
Inner avenue	0.072	1.96**	0.175	3.71***	0.102	2.52**
Inner district × resident	0.793	16.96***	0.785	16.41***	0.778	15.91***
Streetscape features						
Building pixel	−0.012	−3.80***				
Green pixel	0.012	3.27***				
Road pixel	−0.009	−2.59***				
Sky pixel	0.009	2.34**				
Dissimilarity (Euclidean distance)			−0.006	−4.76***		
Cluster 1					0.212	5.07***
Cluster 2					−0.382	−10.77***
Cluster 3					−0.183	−6.15***
Sample size	36,550		36,550		36,550	
$LL(\widehat{\phi})$	−16,338.25		−16,490.76		−16,415.20	
$\bar{\rho}^2$	0.355		0.349		0.352	

Confidence level of statistical significance: ***: $p \leq 0.01$; **: $p \in (0.01, 0.05]$; *: $p \in (0.05, 0.1]$.

neighborhood. Table 6 summarizes the elasticities of pixel variables for each neighborhood. In both neighborhoods, the Green pixel value is positively associated with street perception probability. The elasticities suggest a non-trivial effect size, and the confidence intervals are also positive. The Sky pixel value is also positively associated with perception probability in Kiba, within the entire range of the confidence interval. The Building pixel value may have a non-trivial impact in terms of effect size, but the effect can be negative or positive. The means of the elasticities are positive and negative in Kiyosumi-Shirakawa and Kiba, respectively, but the confidence intervals include both negative and positive values in both neighborhoods. This is the same as the Road pixel variable, although the mean of its elasticity is considerable in Kiyosumi-Shirakawa.

4.3. Scenario evaluation

We then evaluated several policy scenarios by simulation using the estimated model (Model 5). Two scenarios were considered: (1) weak-

Table 4
Estimation results of Model 5 with segmentation by neighborhood.

	Kiyosumi-Shirakawa		Kiba	
	Estimate	t-test	Estimate	t-test
Intercept	−1.454	−5.45***	1.260	18.41***
Distance (km)	−6.630	−21.50***	−4.815	−37.76***
Distance × demographic chars.				
40 yrs or older	−0.269	−3.36***	−0.210	−5.88***
Female	−0.052	−1.52	−0.588	−13.50***
Live less than 10 yrs	1.103	12.30***	0.727	16.09***
Visit once a month or less	0.423	3.84***	0.189	3.78***
Distance × direction				
East	2.877	10.97***	1.329	10.68***
North	2.611	7.09***	3.230	27.68***
South	3.882	13.11***	1.988	12.75***
Boundaries				
Inner river	2.335	22.65***	0.730	10.70***
Inner avenue	0.612	6.52***	0.217	2.57**
Inner district × resident	0.691	9.15***	0.480	7.21***
Streetscape features				
Building pixel	0.005	2.68***	−0.005	−4.51***
Green pixel	0.025	5.48***	0.011	5.59***
Road pixel	0.020	3.75***	−0.001	−2.78***
Sky pixel	0.001	8.21***	0.010	3.86***
Sample size	12,900		23,650	
$LL(\widehat{\phi})$	−4714.45		−10,254.79	
$\bar{\rho}^2$	0.471		0.374	

Confidence level of statistical significance: ***: $p \leq 0.01$; **: $p \in (0.01, 0.05]$; *: $p \leq (0.05, 0.1]$.

Table 5
Effect magnitude of variables affecting neighborhood cognition at the individual level: means and 95 % confidence intervals of direct marginal effect.

	Inner river	Inner avenue	Inner district
Kiyosumi-Shirakawa	0.281	0.074	0.083
	[0.252, 0.310]	[0.057, 0.093]	[0.069, 0.102]
Kiba	0.110	0.033	0.072
	[0.091, 0.125]	[0.016, 0.051]	[0.056, 0.087]

ening the presence of the avenue boundary and (2) increasing in Green pixel values of streets. Among the three boundary variables, weakening the presence of avenue boundaries can be considered to be the most realistic policy, and it may be achieved in several ways, such as by reducing traffic volume or reallocating street space from car- to pedestrian-centered. The Green pixel was positively associated with perception probability in both neighborhoods. Koto Ward, in which the neighborhoods are located, also promotes the idea of "City in the Green" in its urban development policy (Koto City, 2023). Therefore, we consider a scenario of increasing the Green pixel values, where we decreased the Building pixel values (and the values of other pixels if they were insufficient) so that, for each street, the total pixel values is 100, and each pixel value is within the range of $[0, 100]$. This policy of increasing Green pixel values includes, for example, planting trees on a street or greening the walls of buildings along the street. Table 7 summarizes the changes in street perception probability for different scenarios. The confidence intervals were estimated using the bootstrapping method with 100 iterations. The weakening the presence of the avenue boundary increases the perception probability in approximately 4 % on average in Kiyosumi-Shirakawa, but it does not have a significant impact in Kiba. In contrast, the increase in visible green on streets may have a non-trivial impact on perception probability. In Scenario 5, in which the Green pixel value increases by 30 and the avenue boundary is weakened, the mean perception probabilities are 40.5 % and 35.9 % in Kiyosumi-Shirakawa and Kiba, respectively, which are 11.1 % and 7.9 % higher compared to the base scenario.

Fig. 7 compares the mean perception probabilities of the streets on maps between the base scenario and Scenario 5. In Kiyosumi-Shirakawa, the cognitive area is mainly extended to the east by implementing policies, and the river boundary still seems to function as a strong edge of neighborhood cognition. In Kiba, the perception probabilities of the streets in the area seem to increase slightly, regardless of their locations.

4.4. Inter-individual heterogeneity analysis

We also performed an analysis of inter-individual heterogeneity in neighborhood cognition, particularly focusing on familiarity with neighborhoods. In this analysis, using the two familiarity measures of the residential years and visit frequency, as well as the age variable, we evaluated how the street perception probability for each group of respondents changed according to distance. For simplicity, we fixed the street spatial attributes, except for distance, as follows: a street is assumed to be located in the northeast (i.e., the north and east variables are 0.5 each), inside the river and avenue boundaries but outside the address-based neighborhood, and have the same pixel values as the neighborhood average. We herein considered a woman in the interest of brevity and computed her perception probability of the street for different distances in the range of [0.00, 2.00] in kilometers. This was iterated for different characteristics for age and familiarity measures.

Fig. 8 shows the results. Consistent with the findings in Sections 3.3 and 4.2, the more familiar people are with the neighborhoods, the more

Table 6
Effect magnitude of variables affecting neighborhood cognition at the individual level: means and 95 % confidence intervals of direct elasticities.

	Distance	Building pixel	Road pixel	Sky pixel	Green pixel
Kiyosumi-Shirakawa	−2.090	0.154	0.535	0.005	0.106
	[−2.294, −1.875]	[−0.372, 0.536]	[−0.018, 0.912]	[−0.129, 0.120]	[0.026, 0.173]
Kiba	−2.231	−0.160	−0.025	0.079	0.060
	[−2.377, −2.099]	[−0.347, 0.073]	[−0.243, 0.194]	[0.023, 0.143]	[0.023, 0.101]

Table 7
Policy analysis for weakening avenue barrier and increasing street greenery: changes in street perception probabilities with 95 % confidence intervals.

	Base	Scenario 1	Scenario 2	Scenario 3	Scenario 4	Scenario 5
Weaken avenue barrier	x	✓	x	x	✓	✓
Increase Green pixel	x	x	+ 20	+ 30	+ 20	+ 30
Kiyosumi-Shirakawa	0.294	0.333	0.342	0.365	0.382	0.405
	[0.275, 0.314]	[0.310, 0.357]	[0.317, 0.367]	[0.338, 0.398]	[0.354, 0.409]	[0.376, 0.437]
Kiba	0.280	0.287	0.329	0.352	0.336	0.359
	[0.265, 0.297]	[0.269, 0.305]	[0.310, 0.349]	[0.329, 0.373]	[0.316, 0.357]	[0.335, 0.381]



Fig. 7. Change in street perception probabilities by policy.

sensitive to distance they are, hence the faster the drop in perception probability with respect to distance. This trend is particularly clear for residents. For example, let us focus on the distance at which the perception probability becomes 50 % and its difference between two groups of residents (1) who are younger than 40 years and lived in the area for less than 10 years (light-blue lines in Fig. 8a, c), and (2) who are 40 years or older and lived for 10 years or more (red lines). The difference is approximately 325 m in Kiyosumi-Shirakawa and 250 m in Kiba, respectively. In contrast, the difference in perception probability is less obvious for visitors, although those who visit more frequently are more sensitive to distance in neighborhood cognition.

Although not consistent with our original hypothesis, we consider these results to be reasonable. For residents, their cognitive area shrinks to become close to the address-based neighborhood as they live longer in the neighborhood. Therefore, the results may explain a process of certainty acquisition of the spatial area image for residents. For visitors, our results may reflect a search and optimization process of the spatial sphere for their activities. People who do not visit the neighborhood very often may search the places in which their activities occur, resulting in vague neighborhood cognition, whereas those who visit frequently are familiar with the neighborhood, and the places for their activities gradually become optimized.

5. Discussion

Our results are consistent with the findings in the literature, such as boundary effects, sensitivity of women and older people to distance (e.g., Guimpert & Hurtubia, 2018), and positive impacts of the presence of visible greenery on street perception (e.g., Wang et al., 2022). Consistent results from different cities in different countries can reinforce the policy implications. The present study also contributed several new findings to the literature, including the negative association of neighborhood familiarity with the extent of cognitive area. As such, inter-individual heterogeneity exists in the spatial image of a city, and capturing it improves the interpretability of results.

The lack of scalability and generalizability of the results represents a major limitation of the study, as with previous interview-based research on neighborhood cognition (e.g., Guimpert & Hurtubia, 2018; Lynch, 1960; Smith et al., 2010). While this study manually performed street photography and landscape feature extensions for the accuracy of analysis, these processes can be automated by using open data of street images and machine learning techniques (e.g., semantic segmentation). As in some recent studies (e.g., Filomena et al., 2019; Wang et al., 2022), making use of such open datasets and algorithms could extend the applicability of the study in different urban contexts and improve the generalizability of results.

The parametric modeling of sensitivity to distance, including interpersonal heterogeneity, can contribute to research on cognitive distance (Manley et al., 2021). We also incorporated directional heterogeneity and captured non-concentric extent of a cognitive area. However, the literature found that the shape of a cognitive map can be defined by multiple landmarks (Sadalla et al., 1980) or anchor points (Couclelis et al., 1987). From this perspective, the selection of the observation point (reference point) in this study may have a non-trivial impact on the

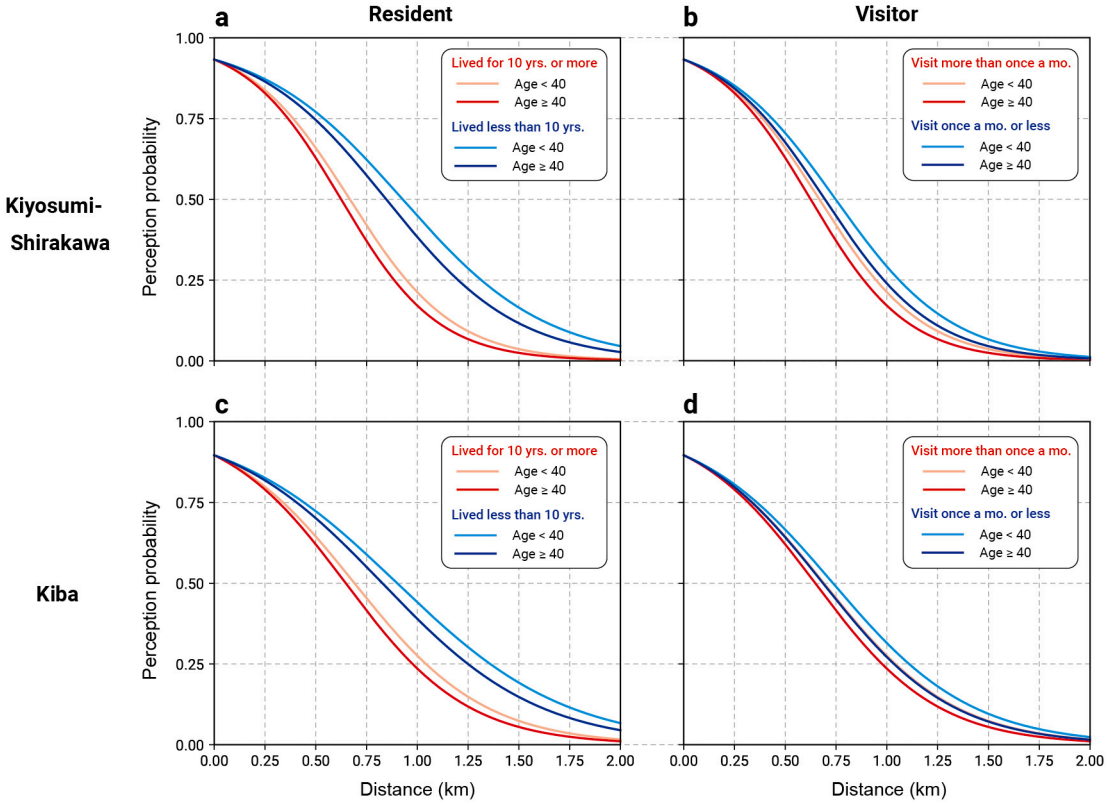


Fig. 8. Perception probability for different distances: Residents and visitors in Kiyosumi-Shirakawa (a–b); and in Kiba (c–d). Red lines show the results for individuals who are more familiar with the neighborhoods, and blue lines show those for individuals who are less familiar. Darker colors indicate older people. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

results. The inclusion of multiple reference points and their definition for community is thus an important step for future work. We found that people who are more familiar with the neighborhoods tend to have a narrower cognitive area. This opens up an extended research question of how an individual's cognitive area changes over time and experience. However, given that the cognitive areas of residents who have lived in Kiyosumi-Shirakawa for more than 10 years tended to be close to the address-based neighborhood, the future study should also investigate the case in which the influence of address-based neighborhood is excluded.

Additionally, the street-based neighborhood cognition model allowed for quantitative evaluation of some policy scenarios. Because the model predicts the perception probability of streets, it also has the potential to be used as a choice set formation model for individuals performing activities in city centers. A limitation related to modeling is that our model predicted the perception probability of each street independently, although considering its spatial attributes. The consideration of the spatial dependence of the utilities (Cliff & Ord, 1981) can significantly improve the model prediction performance and policy evaluation. Modeling spatial correlation is a main subject in spatial econometrics or spatial statistics, and this has been generally applied to spatial data (Anselin, 1988; LeSage & Pace, 2009). To consider the heterogeneity of individuals who perceive spaces in addition to the autocorrelation of spatial data (e.g., Sener et al., 2011), methodological developments may be required, which is an important direction for future studies.

6. Conclusion

This study performed a street-based neighborhood cognition analysis based on the data collected from maps drawn by respondents and street photography in two adjacent neighborhoods of Kiyosumi-Shirakawa and Kiba, Tokyo, Japan. Developing an individual- and street-based perception model, we obtained the following main results and policy implications.

First, the distance and the existence of boundaries are significant factors affecting the probability that a street is perceived within the spatial image of a respondent. The further a street is located from the reference point, the less likely it is to be perceived. Among the boundaries, water boundaries play as the strongest edge of neighborhood cognition in both neighborhoods. The estimated effect sizes and simulation results showed that mitigation of boundaries could contribute to expanding the spatial images of the study neighborhoods.

Second, the landscape features of streets significantly affect the perception of the respondents. In particular, the presence of visible greenery on streets is positively associated with the perception probability and has a non-trivial effect, as suggested by the estimation results with respect to the pixel values extracted from street images. The simulation results suggested the possibility of significantly expanding the cognitive area of the respondents by increasing visible greenery on the streets of the two neighborhoods; thereby, streetscape designs can play an important role in enhancing the spatial image. Note that the revealed difference in effect sizes between the neighborhoods suggests the need for streetscape designs according to the existing neighborhood

images.

Third, our results revealed interpersonal heterogeneity in neighborhood cognition, focusing on the sensitivity to distance. Interestingly, people who are more familiar with the neighborhoods are more sensitive to distance, meaning that their cognitive area is more spatially restrictive than that of those who are less familiar. Although this trend was observed for both residents and visitors, it was clearer for residents. Residents who have lived longer in the neighborhood have acquired more certainty with respect to the spatial image, and their cognitive area forms similarly to the address-based neighborhood.

CRediT authorship contribution statement

Yuki Oyama: Writing – original draft, Visualization, Software, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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Appendix A. Examples of cognitive maps

Fig. A.9 shows examples of cognitive neighborhood areas drawn by respondents, where the basemap is the same as the one used in the survey. People perceive the areas of the neighborhoods differently, but the main parks (Kiyosumi Garden & Park and Kiba Park) were included in the cognitive areas of almost all respondents. Particularly in Kiyosumi-Shirakawa (Fig. A.9a), all respondents included the main park. Most residents drew their cognitive areas that did not cross over the southern part of the river boundary (as the red, green and purple polygons). In Kiba (Fig. A.9b), most visitors included the entire area of Kiba Park within their cognitive areas (as the green and purple polygons). In contrast, some residents included Kiba Park only partially (as the red and blue polygons).

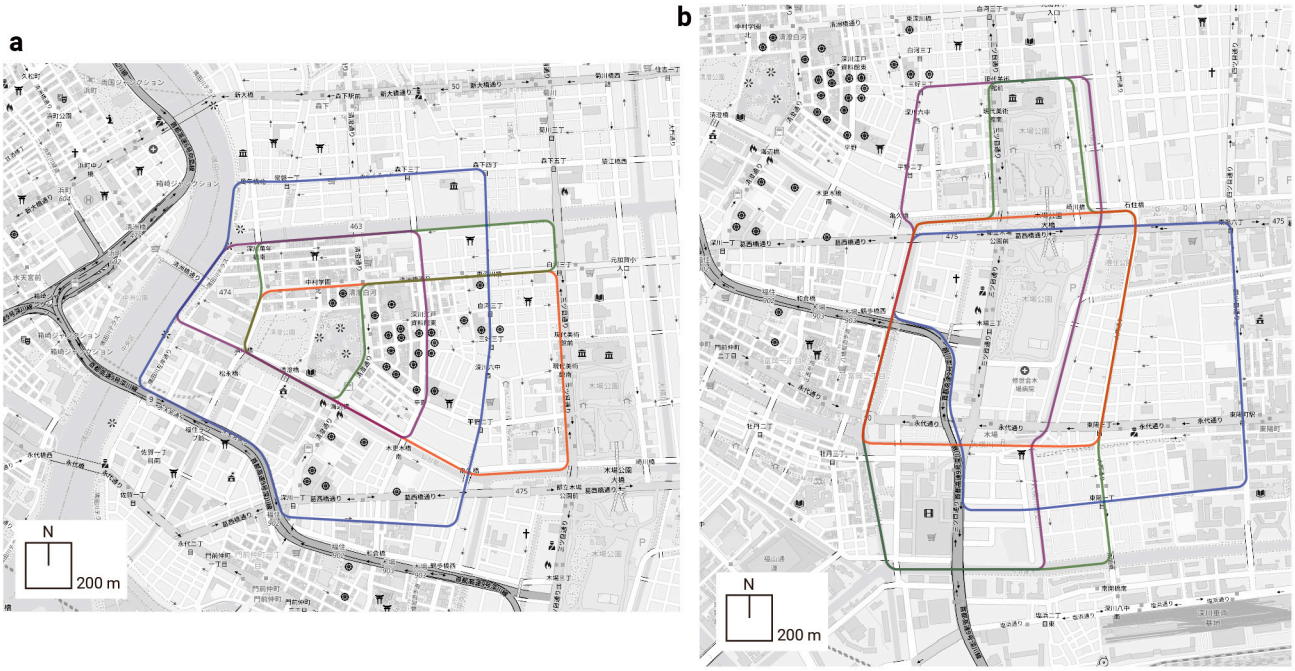


Fig. A.9. Examples of cognitive areas drawn by four respondents, indicated with different colors, for each neighborhood of (a) Kiyosumi-Shirakawa and (b) Kiba. (Basemap: OpenStreetMap).

Appendix B. Direction variable definition

We define direction variables based on the azimuth of the location of each street from the reference point (x in Fig. B.10). For the aggregation in Section 3.1, we simply categorize the directions of the streets into west if $-135 \leq x < -45$, north if $-45 \leq x < 45$, east if $45 \leq x < 135$, and south if $135 \leq x < 180$ or $-180 \leq x < -135$.

For model analysis, we define the directions as continuous variables as follows:

$$(\text{North}, \text{East}, \text{South}, \text{West}) = \begin{cases} \left(0, 0, \frac{180+x}{90}, \frac{-90-x}{90}\right) & -180 \le x < -90 \\ \left(\frac{90+x}{90}, 0, 0, \frac{-x}{90}\right) & -90 \le x < 0 \\ \left(\frac{90-x}{90}, \frac{x}{90}, 0, 0\right) & 0 \le x < 90 \\ \left(0, \frac{180-x}{90}, \frac{x-90}{90}, 0\right) & 90 \le x < 180 \end{cases} \tag{6}$$

The example in Fig. B.10 shows a case of $0 \le x < 90$.

Note that the definition of distance and street photography relied on the entrance (closest node to the reference point) of the street to consider the access from the network. In contrast, we used the center of the street to define the direction variable as it represents the positional relation between the reference point and a street regardless of the network.

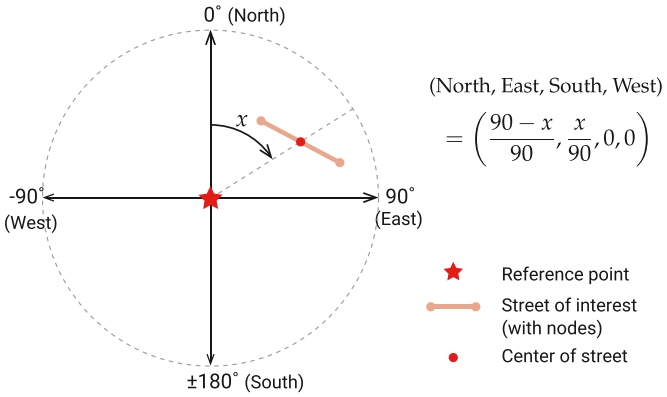


Fig. B.10. Definition of direction variables, where x is the azimuth of the street. This figure shows an example of a street located in the first quadrant (the northeast) based on the reference point.

Appendix C. Street images for different clusters

Fig. C.11 shows, for example, images of streets classified into different clusters by clustering analysis in Section 3.4. The images in Cluster 1 include a large portion of visual greenery. The streets in Cluster 2 tend to have narrow roads, and buildings occupy a large part of their images. The streets in Cluster 3 have similar streetscape but a smaller portion of buildings compared with those in Cluster 2. Cluster 4 seem to include streets with wide roads and a sky view.

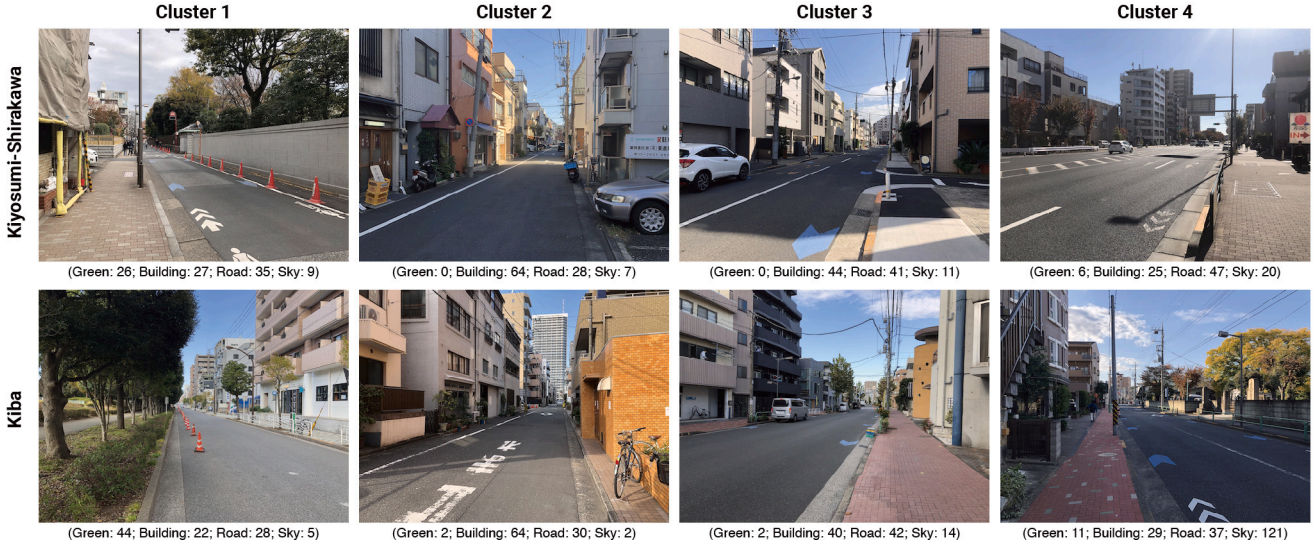


Fig. C.11. Street images for different clusters. An image for each cluster for each neighborhood is shown for example. All photo were taken from the street photography performed for this study.

Appendix D. Model transferability

To analyze the heterogeneity in neighborhood cognition between the two neighborhoods, we performed validation of the model transferability. Because directional effects are clearly site-specific, we used the utility specification of Models 1 and 4, which introduced the pixel values as the streetscape feature input, for this validation. We generated 10 samples of the set of 2000 observations randomly sampled from each of Kiyosumi-Shirakawa, Kiba, and their composition (All) dataset. For each sample, we calculated the indicators for model transferability, and Table D.8 reports the averages over 10 samples. Four indicators were calculated: ACC (accuracy), the percentage of correct prediction based on the highest probability; mean log-likelihood $\bar{LL}_{\mathrm{V}}(\widehat{\boldsymbol{\phi}}_{\mathrm{E}})/|\mathscr{D}_{\mathrm{V}}|$ of the model applied to the validation sample $\mathscr{D}_{\mathrm{V}}$ with the parameters $\widehat{\boldsymbol{\phi}}_{\mathrm{E}}$ estimated with the estimation sample $\mathscr{D}_{\mathrm{E}}$; as well as the following two indexes, transfer rho-square (Tρ^2) and transfer index (TI), defined as (see Parady et al., 2021, for more details):

$$T\rho^2_{E\to V} = 1 - \frac{LL_V(\widehat{\boldsymbol{\phi}}_E)}{LL_V(\widehat{v}_0)} \tag{D.1}$$

$$TI_{E\to V} = \frac{LL_V(\widehat{\boldsymbol{\phi}}_E) - LL_V(\widehat{v}_0)}{LL_V(\widehat{\boldsymbol{\phi}}_V) - LL_V(\widehat{v}_0)} \tag{D.2}$$

The results of Tρ^2 and TI suggest clear heterogeneity in neighborhood cognition between the two neighborhoods; particularly, the transfer from the model estimated with the Kiyosumi-Shirakawa data to that of Kiba was not successful and even obtained a worse result than a model with only the intercept. This may be because the sensitivity to distance and the effect of the river boundary were large in Kiyosumi-Shirakawa, and the model underestimated the perception probability of Kiba where the cognitive area and network size were relatively large.

Table D.8
Validation of model transferability. The reported values are the averages over 10 iterations of validation.

Estimation sample	Validation sample	ACC	$\bar{LL}(\widehat{\phi}_{\mathrm{E}}, \mathscr{D}_{\mathrm{V}})$	Tρ^2	TI
All	All	0.815	−0.437	0.257	1.000
All	Kiyosumi-Shirakawa	0.834	−0.401	0.340	0.864
All	Kiba	0.795	−0.473	0.165	0.860
Kiyosumi-Shirakawa	All	0.813	−0.497	0.156	0.602
Kiyosumi-Shirakawa	Kiyosumi-Shirakawa	0.838	−0.369	0.393	1.000
Kiyosumi-Shirakawa	Kiba	0.789	−0.625	−0.103	−0.559
Kiba	All	0.805	−0.451	0.234	0.909
Kiba	Kiyosumi-Shirakawa	0.812	−0.444	0.270	0.687
Kiba	Kiba	0.798	−0.458	0.191	1.000

Appendix E. Original questionnaire

See Fig. E.12.

清澄白河エリア *Kiyosumi-Shirakawa Area*

① あなたが清澄白河エリアだと思う範囲を地図上にどのような形でも大きさでも良いので囲ってください。

Please enclose on the map in any shape and size the area you consider to be the Kiyosumi-Shirakawa area.



② 以下の当てはまる項目にチェック✓をお付けください。

Please check ✓ for each of the following items that apply to you.

●性別 *Gender*

□男性 *Male* □女性 *Female*

●年齢 *Age*

□10 代 *10s* □20 代 *20s* □30 代 *30s* □40 代 *40s* □50 代 *50s* □60 代 *60s* □70 代以上 *70s or older*

●居住年数 *Year of residence*

□1 年未満 *Less than 1 year* □1 年以上 5 年未満 *1 year to less than 5 years* □5 年以上 10 年未満 *5 years to less than 10 years* □10 年以上 20 年未満 *10 years to less than 20 years* □20 年以上 *20 years or more*

●清澄白河を訪れる頻度 *Frequency of visiting Kiyosumi-Shirakawa*

□ほとんど毎日 *Almost everyday* □週 4~5 日 *4-5 days a week* □週 2~3 日 *2-3 days a week* □週 1 日 *1 day a week* □月 2~3 回 *2-3 times a month* □月 1 回 *Once a month* □それ以下 *Less than that*

Fig. E.12. Questionnaire used in the neighborhood cognition survey. The English translation is by DeepL and was not included at the time of the survey.

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